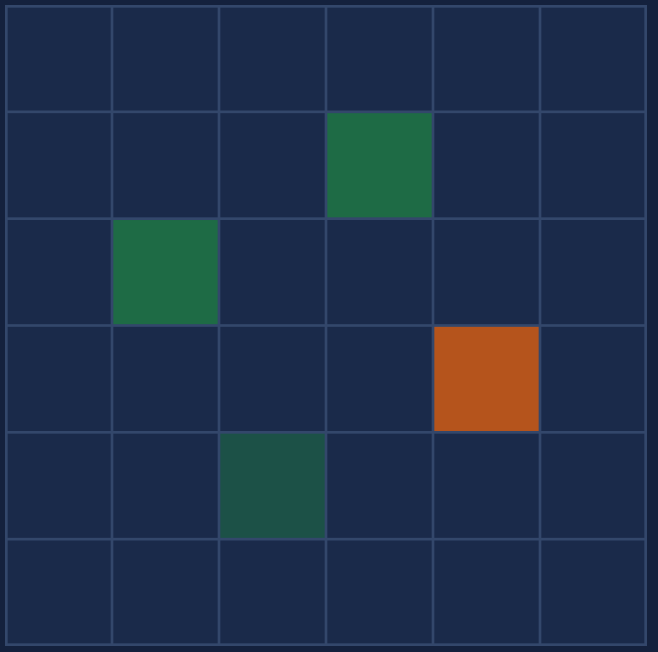
INF131 · WEEK 4 · LAB GUIDE

Cleaning data with Power Query

Step by step, one click at a time. No Excel or programming experience needed.



BEFORE YOU START

What you need and what you will make

Excel on a PC

Microsoft 365 Excel on Windows, like the lab computers. The browser version of Excel can’t do all the steps.

The data file

inflation_raw.xlsx from World Bank. Save it in a folder you can find again.

What you make

inflation_clean.xlsx: a clean table, plus the list of steps you used.

THE 9 STEPS

1 Keep raw file · 2 Import · 3 Open editor · 4 Headers · 5 Replace “..” · 6 Remove notes · 7 Remove duplicates · 8 Data types · 9 Load and save

BEFORE YOU START

Seven words you will see

Word	What it means
Workbook	An Excel file.
Sheet	A page inside the workbook. The tabs at the bottom of the window.
Ribbon and tabs	The menu at the top: File, Home, Insert, Data… Click a tab to see its buttons.
Column header	The name at the top of a column, such as Country Name.
Power Query Editor	A separate window where you clean data before it goes into Excel.
Applied Steps	The list of everything you did, on the right of the editor.
null	Power Query’s word for “empty, no value here”.

BEFORE YOU START

Inside the file: inflation in 8 countries

World Bank, World Development Indicators · World countries inflation data from 1960 to 2025. Many values will be missing because records were not kept or uploaded, so we have to deal with them

1	Data Source	World Development Indicators									
2	Last Updated Date	10/8/2026									
3											
4	Country Name	Country Code	Indicator Name	Indicator Code	1960	1961	1962	1963	1964	1965	1966
5	Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
6	Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
7	Afghanistan	AFG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
8	Africa Western and Central	AFW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
9	Angola	AGO	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
10	Albania	ALB	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
11	Andorra	AND	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
12	Arab World	ARB	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
13	United Arab Emirates	ARE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
14	Argentina	ARG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
15	Armenia	ARM	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
16	American Samoa	ASM	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
17	Antigua and Barbuda	ATG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
18	Australia	AUS	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	3.728821504	2.287580407	-0.31948864	0.641025286	2.866226699	3.405584767	3.293411469
19	Austria	AUT	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	1.945749352	3.542239867	4.381798639	2.708766995	3.868563549	4.930916377	2.054820593
20	Azerbaijan	AZE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							
21	Burundi	BDI	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG							4.436860068

".." = no data · a repeated row · empty rows and notes at the bottom · #.# = a number (yours will show real values)

STEP 1 OF 9

Keep the raw file safe

`File` › `Save As`

1. Download **inflation_raw.xlsx** from website.
2. Make a folder, for example Documents › INF131 › Week4, and put the file there.
3. Do not open or change this file. Power Query will read it for you.
4. Open Excel and click **Blank workbook**.
5. Save it right away as **inflation_clean.xlsx** in the same folder.

1. Download directly on Power Query using this link too: https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG?locations=AL
2. Carry out all the steps and save a clean file
3. Save it right away as **inflation_clean.xlsx** in the same folder.

YOU SHOULD SEE

Two files in your folder:

`inflation_raw.xlsx`
`inflation_clean.xlsx`

The raw file never changes. All your work goes into the clean file.

STEP 2 OF 9

Import the file

`Data` › `Get Data` › `From File` › `From Excel Workbook`

1. In a new excel blank notebook, click **Get Data**, on the far left of the ribbon.
2. Point to **From File**, then click **From Excel Workbook**.
3. Find **inflation_raw.xlsx** in your folder, click it once, then click **Import**.
4. We can also get data from Database directly using the link

YOU SHOULD SEE

A new window called **Navigator**.

It may take a few seconds to appear. Don’t click twice.

STEP 3 OF 9

Open the Power Query Editor

`Navigator` › `Data` › `Transform Data`

1. On the left of the Navigator, click the sheet called **Data**.
2. A preview of the table appears on the right.
3. At the bottom, click **Transform Data**.

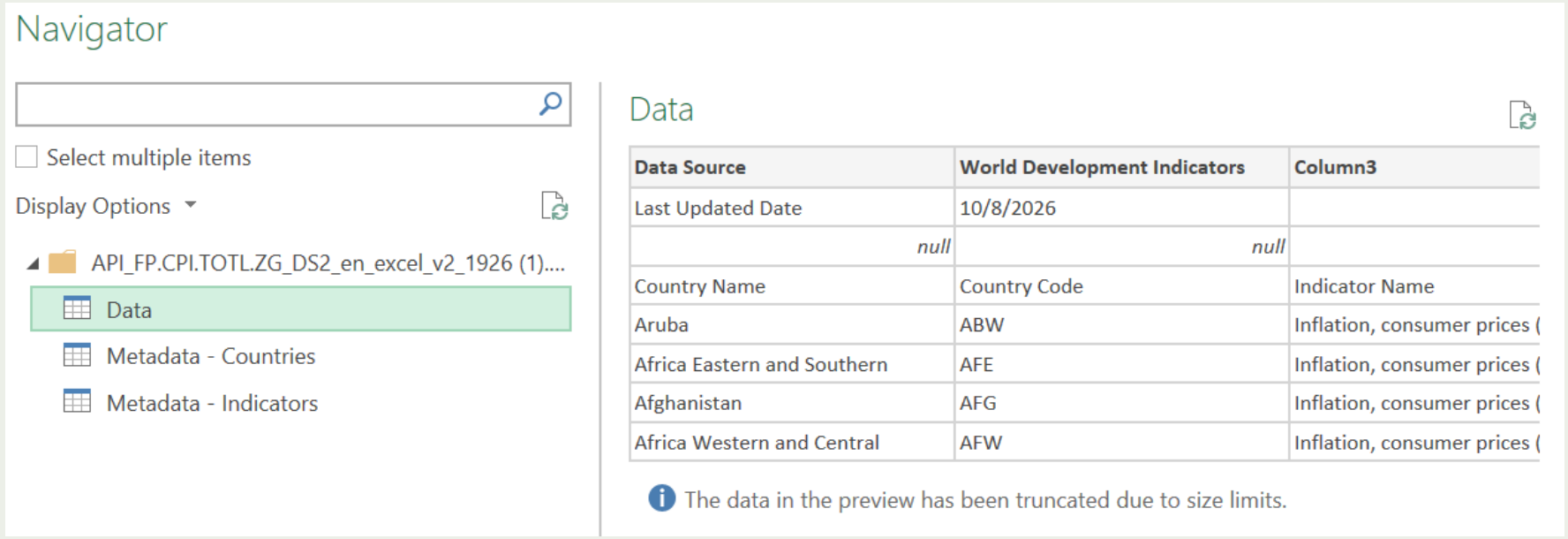
⚠ Click **Transform Data**, not Load. Load skips the cleaning.

YOU SHOULD SEE

A new, larger window: the **Power Query Editor**. The next page shows what is in it.

A QUICK TOUR

The Power Query Editor



1 Tabs. Home, Transform and View hold every button you need.

2 Your data. The icon before each name shows its type: ABC = text, 1.2 = number.

3 Applied Steps. Every click is saved here. Click × on a step to undo it.

4 Row count, bottom left. Watch it change.

Tip: click **View** and tick **Column quality**. Each column then shows how many cells are Valid, Error or Empty.

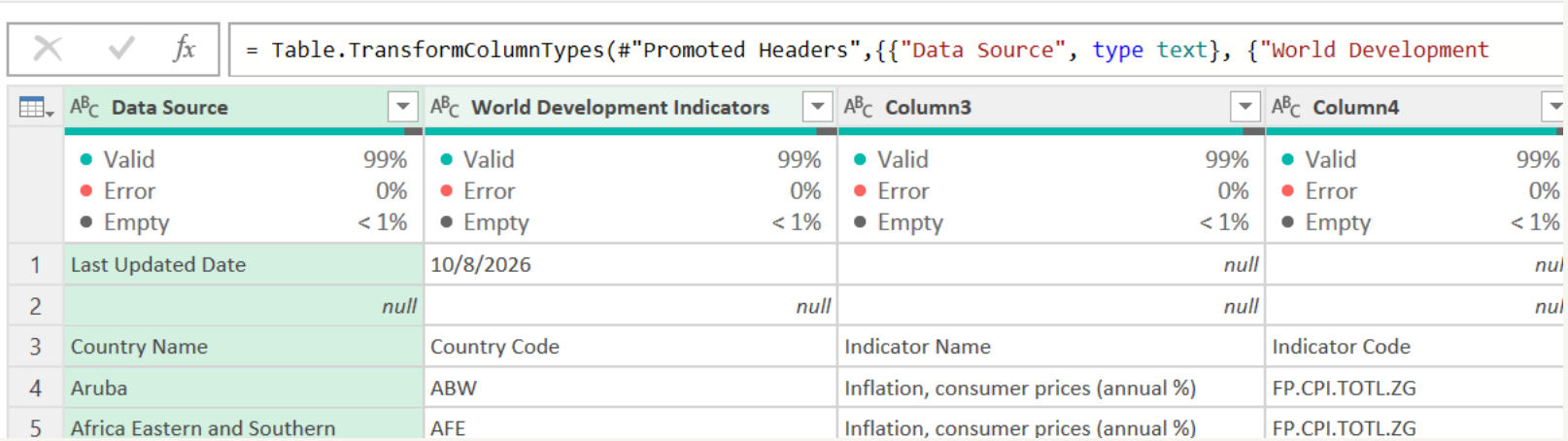
STEP 4 OF 9

Check the column names

`Home` › `Use First Row as Headers`

1. Look at the top of each column.
2. Names like **Country Name** and **Country Code**? Good: Power Query already did this step.

   Go to Step 5.
3. Names like **Column1, Column2**, with the real names in the first row? Click **Home › Use First Row as Headers after removing 2 rows**



	Data Source	World Development Indicators	Column3	Column4
	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%	Valid 99% Error 0% Empty < 1%
1	Last Updated Date	10/8/2026	*null*	*nul*
2	*null*	*null*	*null*	*nul*
3	Country Name	Country Code	Indicator Name	Indicator Code
4	Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG
5	Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG

YOU SHOULD SEE

```
Country Name
Country Code
Series Name
Series Code
2015 [YR2015]
…
2024 [YR2024]
```

In Applied Steps: **Promoted Headers**

STEP 5 OF 9

Replace “..” with null

`Transform` › `Replace Values`

1. Click the header **2015 [YR1992]**.
2. Scroll right. Hold **Shift** and click the header **2024 [YR2024]**. All year columns are now selected.
3. Click the **Transform** tab, then **Replace Values**.
4. Value To Find: type **..** (two dots).
5. Replace With: type **null** in small letters. Click **OK**.

THE BOX LOOKS LIKE THIS

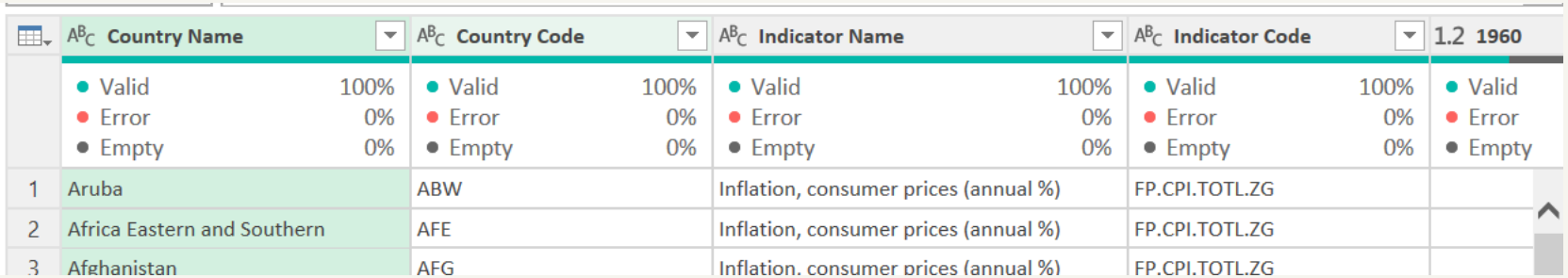
Value To Find

..

Replace With

null

Afterwards, cells that said .. now say *null*. New step: **Replaced Value**.



	Country Name	Country Code	Indicator Name	Indicator Code	1960
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	Aruba	ABW	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	
2	Africa Eastern and Southern	AFE	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	
3	Afghanistan	AFG	Inflation, consumer prices (annual %)	FP.CPI.TOTL.ZG	

STEP 6 OF 9

Remove empty rows and notes

A · EMPTY ROWS

Home › Remove Rows › Remove Blank Rows

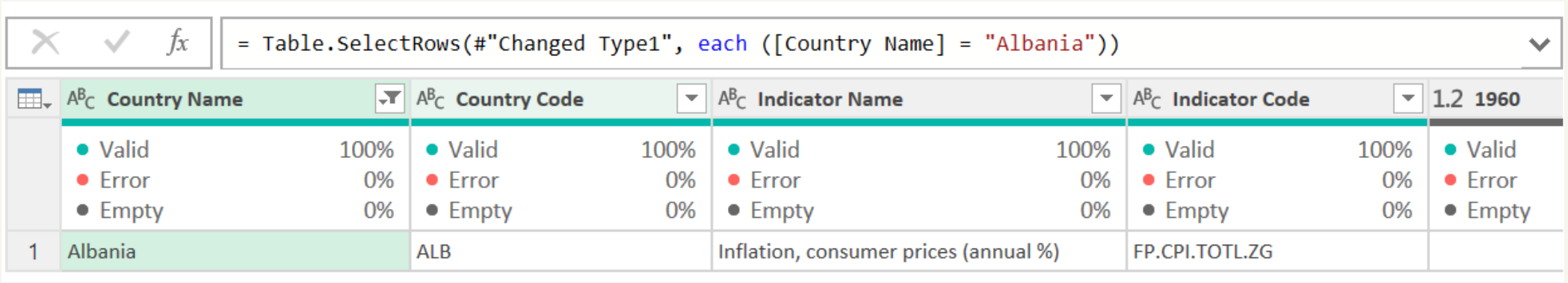
B · NOTE ROWS AT THE BOTTOM

1. Click the small arrow at the right of the **Country Code** header.
2. Unselect all and keep Albania only.
3. Click **OK**.

YOU SHOULD SEE

The table ends with the last country. The rows “Data from database…” and “Last Updated…” are gone.

New steps: **Removed Blank Rows** and **Filtered Rows**. The row count went down.



STEP 7 OF 9

Remove duplicate rows

`Ctrl + A` › `Home` › `Remove Rows` › `Remove Duplicates`

1. Click the header **Country Name**.
2. Right click -> Unpivot Other Columns.
3. Click **Home › Remove Rows › Remove Duplicates**.

	Country Name	Attribute	Value
	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%
1	Albania	Country Code	ALB
2	Albania	Indicator Name	Inflation, consumer prices (annual %)
3	Albania	Indicator Code	FP.CPI.TOTL.ZG
4	Albania	1992	*226.0054213*
5	Albania	1993	*85.00475124*
6	Albania	1994	*22.56505269*
7	Albania	1995	*7.793218538*
8	Albania	1996	*12.72547781*
9	Albania	1997	*33.18027438*



STEP 9 OF 9

Load the clean table and save

Home › Close & Load

1. In the editor, click **Home › Close & Load**.
2. The editor closes and Excel adds a new sheet with your table.
3. Double-click the sheet tab at the bottom and rename it **Inflation**.

	Country Name	Attribute	Value
	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%
1	Albania	1992	226.0054213
2	Albania	1993	85.00475124
3	Albania	1994	22.56505269
4	Albania	1995	7.793218538
5	Albania	1996	12.72547781
6	Albania	1997	33.18027438
7	Albania	1998	20.64285887
8	Albania	1999	0.389437654
9	Albania	2000	0.050018136
10	Albania	2001	3.10758827
11	Albania	2002	7.770525834
12	Albania	2003	0.484002612

YOU SHOULD SEE

A striped table with 8 countries.

On the right, the **Queries & Connections** panel: **8 rows loaded**.

	Country Name	Attribute	Value
2	Albania	1992	226.0054213
3	Albania	1993	85.00475124
4	Albania	1994	22.56505269
5	Albania	1995	7.793218538
6	Albania	1996	12.72547781
7	Albania	1997	33.18027438
8	Albania	1998	20.64285887
9	Albania	1999	0.389437654
10	Albania	2000	0.050018136
11	Albania	2001	3.10758827

ONE-PAGE SUMMARY · PRINT THIS

All nine steps

#	Step	Where to click
1	Keep the raw file	Save inflation_raw.xlsx in a folder and never edit it. New blank workbook › save as inflation_clean.xlsx
2	Import	Data › Get Data › From File › From Excel Workbook › choose the file › Import
3	Open the editor	Navigator: click the sheet Data › Transform Data
4	Column names	Only if you see Column1, Column2: Home › Use First Row as Headers
5	Missing values	Select all year columns (click first, Shift + click last) › Transform › Replace Values: .. → null
6	Empty rows and notes	Home › Remove Rows › Remove Blank Rows. Then Country Code arrow › untick (null) › OK
7	Duplicates	Click a header, Ctrl + A › Home › Remove Rows › Remove Duplicates
8	Data types	Names and codes › Text. Year columns › Decimal Number. “Replace current” if asked
9	Load and save	Home › Close & Load › rename the sheet › Ctrl + S

Finished when: 8 rows · no “..” · no notes at the bottom · year columns are numbers · Applied Steps screenshot saved